

Data Handling

Attendance



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Data Collection

process of gathering and acquiring data from various sources to serve as the foundation for analysis, modeling, and decision-making.

- often the first step in the data science workflow.

What is a Dataset?

- a collection of information containing features and instances

Instances - an individual data or observational units

- often represented as the rows of a data table

Features - variable, characteristic that is measured or observed on an instance

- often represented as the columns of a data table

A	B	C	D	E	F
Customer ID	Age	Gender	Item Purchased	Category	Purchase Amou
1	55	Male	Blouse	Clothing	53
2	19	Male	Sweater	Clothing	64
3	50	Male	Jeans	Clothing	73
4	21	Male	Sandals	Footwear	90
5	45	Male	Blouse	Clothing	49
6	46	Male	Sneakers	Footwear	20
7	63	Male	Shirt	Clothing	85
8	27	Male	Shorts	Clothing	34
9	26	Male	Coat	Outerwear	97
10	57	Male	Handbag	Accessories	31
11	53	Male	Shoes	Footwear	34
12	30	Male	Shorts	Clothing	68
13	61	Male	Coat	Outerwear	72
14	65	Male	Dress	Clothing	51
15	64	Male	Coat	Outerwear	53
16	64	Male	Skirt	Clothing	81
17	25	Male	Sunglasses	Accessories	36
18	53	Male	Dress	Clothing	38
19	52	Male	Sweater	Clothing	48
20	66	Male	Pants	Clothing	90
21	21	Male	Pants	Clothing	51
22	31	Male	Pants	Clothing	62

4	3	50	Male
5	4	21	Male
6	5	45	Male



Data Collection cont.

Types of Data:

Structured Data: typically numerical data or quantitative data, and organized in a table

- easy to understand by machine learning algorithms
- Example: Customer details (name, age, purchase history).

Unstructured Data: abstract data or qualitative data, and may or not be organized in a table

- cannot be processed and analyzed through conventional data tools and methods.
- Example: text, images, videos, or audio files.

Semi-structured Data: not purely structured, but also not completely unstructured.

- meaning not being organized in a table
- architecturally contains some level of organization or structure, but does not conform to a rigid schema or data model, and may contain elements that are not easily categorized or classified.
- Example: JSON, XML files, or log data.



cont.

Types of Data Values:

Numerical Values: values having mathematical meaning

- numbers and vectors
- Example:
 - height
 - weight

Categorical Feature: non-numerical values or numerical values without mathematical meaning

- Example:
 - yes/no OR 1/0
 - address

Data for Machine Learning

Types of Machine Learning Models:

Supervised Learning: model is trained on labeled data, meaning the dataset contains both inputs (features) and outputs (labels).

- The goal is for the model to learn the mapping from inputs to outputs so that it can accurately predict the output for new, unseen data.
- supervised learning tasks:
 - Classification
 - regression

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

cont.

Unsupervised Learning: model is trained on data without labeled outputs, meaning the dataset contains only input features (no output labels).

- The goal is to uncover hidden patterns, structures, or relationships in the data.
- unsupervised learning tasks:
 - Clustering
 - Dimensionality Reduction
 - Association
 - outlier detection detection

Records

Age	Gender	Income	Profession	Tenure	City
35	M	60,000	IT	12	KRK
23	F	90,000	Sales	3	WAW
18	M	12,000	Student	1	KRK
42	F	128,000	Doctor	13	KRK
34	M	63,000	Manager	8	WAW
56	M	82,000	Teacher	30	WAW

Preprocessing for Machine Learning



Train-Validation-Test split:

model training on an entire set of sample data carries the risk of overfitting the sample data, leading to poor performance when predicting beyond the sample

sample data is often split into three subsets:

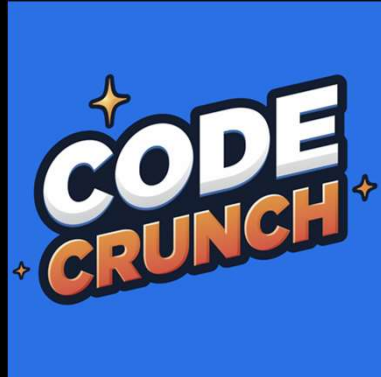
- Training data
- validation data
- Test data

Determining the data split ratio

- Typically you want more training data than validation and testing data
 - 60% - 90% training data
 - equal split between validation and testing data



Q&A Session



Attendance



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Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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